# Behavior-Based Fuel and $CO_2$ Emission Prediction Using Machine Learning Models

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***Abstract*—Accurate fuel consumption and $CO_2$ estimation is of critical importance to achieving a more sustainable future, especially within the transportation and specifically eco-driving sections. In this respect, an estimation framework regarding fuel consumption $CO_2$ emissions based on behavior will specifically be proposed. In this way, various raw parameters related to fuel consumption, such as vehicle speed, various types of acceleration, rotations, throttle movements and engine workload, will specifically be used to calculate various fuel consumption and $CO_2$ emissions prediction parameters, such as harsh acceleration, braking, idle ratio, stop and go, and various types of speed change. In this way, a primary estimation regarding fuel consumption will specifically be followed by a secondary estimation regarding $CO_2$ emissions based on fuel types and specifically fuel consumption. In this respect, there is a primary advantage, i.e., achieving greater transparency that will allow for the inclusion of an 'Explanation Artificial Intelligence' model, i.e., 'SHapley Additive exPlanations' (SHAP), specifically, to attain primary attributions regarding global and local levels related to the framework developed and specifically regarding the estimation accuracy of fuel consumption. Moreover, attestations regarding these achievements will specifically lead to attestations and understandings that fuel consumption estimation is accurate within this framework.**

**Keywords:** Driving behavior analysis, Fuel consumption prediction, $CO_2$ emission estimation, OBD-II and CAN data, Ensemble machine learning; XGBoost regression, Explainable AI (SHAP), Real-world driving data.

## I. Introduction

Road transport pollution emissions were found to account for about one-quarter of the $CO_2$ emissions generated through energy use worldwide [1], [2]. This is currently a more serious issue due to the increasing level of urbanization, car use, and traffic density, and hence the need for the proper use of the fuel, which is imperative for emission control. Conventional calculations of emissions use primarily model calculations such as COPERT, MOVES, and WLTP to calculate the energy consumption and emissions under fixed operational conditions [4], [5]. Although these models are very useful as a benchmark during regulatory tests, these models do not take into account the dynamically changing emissions with respect to fixed operational conditions such as personalized operational conditions like personalized driving behavior, roads, and operational conditions of traffic [6],[7]. Human operational conditions like aggressive acceleration and braking, speed, and idling have fixed impacts on fuel and fixed impacts on the emission of , which were not accounted for by the conventional models of calculation [8]. Thus, although Portable Emissions Measurement Systems provide approaching real-world data measurements, they appear to be too expensive to be used on a scalable basis [9]. Against this background, a large requirement arises to develop low-cost systems that could be easily deployed, behavior-aware, and operable onboard with the help of available infrastructural resources [10]. Recently, machine learning approaches using OBD-II and CAN bus parameters such as vehicle speed, crank angle, engine speed, and mass air flow, have demonstrated their capability in fuel consumption and $CO_2$ emissions estimation [11],[12],[13]. Ensemble learning techniques such as Random Forest and XGBoost have shown the ability to outperform their physics-based model counterparts in a complex operating environment [14],[15]. However, two challenges remain to be overcome: 1) most of those approaches have been developed as a fully offline solution that is not practical yet for a mobile scenario; 2) most of the machine learning models remain a black-box approach, where most of the insights generated are not of value to decision-makers who seek to understand why $CO_2$ emissions deviate according to distinct driving behaviors [16],[17]. To address these shortcomings, in this article, a Real-time Behavior-Driven Estimation System powered by OBD-II and CAN data, which consists of four different functional modules: (i) Advanced Future, which predicts future dynamic behaviors in the road environment, (ii) Fuel Efficiency Level Classifier, which estimates driving behavior for evaluating classification, (iii) OBD-II CAN sensor data driven Emission Model for emissions level predictions, and (iv) Driving Machine Learning for predictions using ensemble models and explainable AI for behavior-driven predictions, will be developed. Experiments will confirm that the intro-

duction of features developed by a behavior-driven estimation system enhances prediction accuracy by a significant extent in the context of sensitivity analysis, according to aggressive driving behavior and complementary traffic-driven behavior. Its real-time capabilities for onboard analysis will also be demonstrated by explanation provided by SHAP explanation in [18],[19],[20].

## II. Literature Review

Machine learning is being increasingly utilized for modeling car fuel and $CO_2$ emission due to its ability to process complex and diverse data. Numerous research works have utilized ML for predicting either fuel consumption or $CO_2$ emission. However, mostly these models were specific to one form of data at a time and failed to explain their findings, failing to consider driving behavior.

Some research uses basic vehicle parameters for emission estimation. Udoh et al. [1] predicted $CO_2$ emission for light-duty vehicles using decision trees and XGBoost algorithms on test datasets. Alam et al. [3] employed deep learning with a technique known as SHAP for explanation on test datasets. Yuan et al. [4] employed another explanation technique known as ALE with XGBoost on test datasets for both normal and hybrid vehicles. In all these research works, actual driving behavior is not taken into consideration and they do not work on real time datasets.

Some other research is more focused on driving behavior. Wang et al. [2] employed the Random Forest model for predicting monthly emissions based on behavior such as aggressive braking. Another research by Du et al. [10] observed driving behavior from GPS and then calculated emissions with the MOVES model. Similarly, research by Jain [11] and Haghshenas et al. [12] employed ODB-II and deep learning for drivers and then predicted fuel and $CO_2$ emissions. However, little research can be found that can integrate GPS, ODB-II, and behavior features. Some studies apply location-based and real-time information. For instance, Alazemi et al. [14] employed the prediction of vehicle emissions from GPS speed profile maps and ML. Chowdappa et al. [7] applied unsupervised learning based on vehicle power. Emissions at busy intersections were assessed by Madziel [5], [6] with sensors and neural networks. However, these do not give a clear explanation for their results and cannot be applied by vehicle drivers. Explainable AI (XAI) is now beginning to emerge. SHAP was utilized by Alam et al. [3] and Wu et al. [17] for explaining the impact of variable inputs on emission production, though this was carried out offline. Ashqar et al. correlated various driver behavior categories with fuel efficiency. However, real-time feedback was not provided. A great majority of current ML approaches fall under the "black box" domain and therefore cannot be readily employed for assisting drivers with eco-driving and driver training.

Some studies adopt a broader framework. Qiu [15] employed deep learning to lower emission production throughout the city by enhancing traffic circulation. Another study by Jeong et al. [24] employed satellite images and a technique known as Gaussian Processes for emission analysis. Gurcan [22] compared various machine learning approaches. Also, another study by Keskin et al. [23] employed LSTM networks with the absence of OBD for predicting fuel consumption.

Several papers reviewed the literature. Mobini Seraji et al. [16] and Al Nuaimi et al. [25] reported several approaches from ML. However, they haven't implemented any model. Another research by Zahid and Jamil [18] employed the number of vehicle-years to forecast the fuel economy with RF. However, they didn't investigate driving behavior. In conclusion, most literature on this subject has either focused on fuel consumption and emission levels simultaneously or on just one aspect and employed only one form of data. Some were not meant for real-time analysis and were unable to explain their forecasts. Some may utilize SHAP and LIME for feature explanation, though they do not really provide sufficient feedback for drivers and insights for transportation planning.

## III. Research Objective

This research work will focus on developing a behavior-driven machine learning estimation system to estimate fuel consumption and carbon emissions, including usage of data derived from onboard OBD-II and CAN bus. The objectives of this research work include the following:

1. Model Development & Multi-Layer Integration: Create and train four different ML models: Advanced Future, Fuel Efficiency Level, OBD-II & CAN Estimation, and Driving ML in order to achieve a description of what is being driven, the estimation of fuel consumption, and the deduction of real-world and CO values.

2. Carried Out Behavioral Consequence: Measure the impact of parameters with a behavioral nature like aggressive acceleration, hard braking, idling, and stop-and-go cycles regarding consumption and $CO_2$ emissions based on feature-derived parameters for a better understanding of driving behavior.

3. Explainability & Interpretability: Use Explainable AI techniques such as SHAP for explaining the reasoning of GPS models at global and local levels for features and routes, respectively.

4. Cross-Dataset Generalization & Robustness: Validate the performance robustness of the proposed models with cross-validation techniques ensemble learning approaches to make sure that the predictions generalize to various settings driven environments models.

5. Real-Time Feasibility & Deployment: The capabilities of real-time processing may be evaluated by inferring their latency and compute overheads in order to deduce their applicability to real-world implementations in vehicles.

## IV. Methodology

This paper discusses the development of the real-time behavior-aware system through vehicle sensors and explainable machine learning algorithms based on estimating fuel and carbon dioxide emissions. This research uses approaches

in ensemble regression learning, structure in regression analysis, behavior change approaches, and model interpretability. The approaches included in this study are data collection, processing, development of features, development of models in regression, ensemble learning, interpretability analysis, and deployment of models.

### A. Regression Models and Explainable AI

*1) Random Forest Regression:* Random Forest (RF) is an ensemble learning technique constructed by aggregating multiple decision trees through bootstrap aggregation (bagging). The final prediction is obtained by averaging the outputs of individual trees, which effectively reduces model variance and improves robustness against noisy sensor data commonly observed in OBD-II measurements.

RF is particularly suitable for fuel consumption and $CO_2$ emission estimation due to its ability to capture nonlinear relationships among vehicular variables such as engine speed (RPM), throttle position, vehicle speed, and acceleration. Additionally, the model can handle feature interactions and irregular driving patterns while resisting overfitting. In this work, RF serves as a baseline regression model to estimate fuel consumption (L/100 km) and $CO_2$ emissions (g/km) under diverse driving conditions.

*2) XGBoost Regression:* Extreme Gradient Boosting (XGBoost) is a gradient-based ensemble learning method in which trees are trained sequentially to minimize prediction error using gradient descent optimization. Unlike Random Forest, XGBoost builds each tree to correct the residual errors of previous trees, resulting in higher predictive accuracy.

XGBoost is well-suited for vehicular emission modeling due to its capacity to learn complex nonlinear dependencies among driving parameters such as speed variance, engine load, throttle opening and RPM. The model also incorporates regularization through L1 and L2 penalties, enhancing generalization performance in real-world driving scenarios including stop-and-go traffic and variable road conditions. Empirical evaluation demonstrates that XGBoost consistently outperforms RF in terms of $R^2$, MAE, and RMSE.

*3) Explainability via SHAP:* Although ensemble models provide high predictive accuracy, they are inherently non-interpretable. To address this limitation, SHapley Additive exPlanations (SHAP) are employed to provide model interpretability. SHAP decomposes each prediction into feature-wise contributions based on cooperative game theory.

Global SHAP analysis identifies dominant behavioral features influencing fuel consumption and emissions, including vehicle speed, RPM response, throttle behavior, idle duration, and aggressive acceleration events. Local SHAP explanations provide trip-level insights, enabling the identification of inefficient driving segments and supporting eco-driving recommendations. This approach bridges the gap between high-performance prediction and transparent decision-making.

### B. Data Acquisition and Preprocessing

The dataset consists of real-world OBD-II sensor readings collected from multiple vehicles and drivers, ensuring ecological validity and robustness. The time-series data include vehicle speed, engine RPM, throttle position, engine load, thermal parameters, and labeled driving styles (Eco, Normal, Aggressive). Baseline fuel economy and $CO_2$ emission data are obtained from the SmartWay Vehicle Dataset.

Preprocessing steps include removal of inconsistent sensor readings, handling of missing data through interpolation and deletion, resampling to a uniform temporal resolution of 1 Hz, and trip segmentation based on ignition status and extended idle periods. These steps transform raw sensor signals into structured driving sessions suitable for regression modeling.

### C. Fuel Consumption and $CO_2$ Emission Conversion

The SmartWay dataset provides values measured in imperial units, which are converted into standard scientific units for modeling. Fuel economy values measured in miles per gallon (MPG) are converted to liters per 100 kilometers (L/100 km) using the following equation:

$$Fuel_{L/100km} = \frac{235.214583}{MPG} \tag{1}$$

TABLE I
FUEL CONSUMPTION CONVERSION PARAMETERS

| Symbol | Meaning |
|---|---|
| $Fuel_{L/100km}$ | Fuel consumption expressed in liters per 100 kilometers |
| $MPG$ | Fuel efficiency measured in miles per gallon (US units) |
| 235.214583 | Fixed conversion constant from MPG to L/100 km |

This formula relates the imperial fuel efficiency measure to standard scientific units amenable to calculation and comparison.

*1) $CO_2$ Emission Conversion Formula:* Carbon dioxide emissions measured in grams per mile are converted to grams per kilometer using:

$$CO2_{g/km} = \frac{CO2_{g/mile}}{1.609344} \tag{2}$$

TABLE II
$CO_2$ EMISSION CONVERSION PARAMETERS

| Symbol | Meaning |
|---|---|
| $CO2_{g/km}$ | Carbon dioxide emissions in grams per kilometer |
| $CO2_{g/mile}$ | Carbon dioxide emissions in grams per mile |
| 1.609344 | Exact mile-to-kilometer conversion factor |

### D. Driving Style-Based Fuel and $CO_2$ Adjustment

Driving behavior effects are incorporated using the driving style label. A style multiplier adjusts baseline vehicle fuel consumption and emission values.

*1) Driving Style-Adjusted Fuel Consumption:*

$$Fuel_{adj} = Fuel_{baseline} \times M_{style} \tag{3}$$

TABLE III
FUEL CONSUMPTION ADJUSTMENT PARAMETERS

| Symbol | Meaning |
|---|---|
| $Fuel_{adj}$ | Driving style-adjusted fuel consumption |
| $Fuel_{baseline}$ | Baseline fuel consumption under normal driving conditions |
| $M_{style}$ | Driving style multiplier |

TABLE IV
$CO_2$ EMISSION ADJUSTMENT PARAMETERS

| Symbol | Meaning |
|---|---|
| $CO2_{adj}$ | Adjusted $CO_2$ emission considering driving behavior |
| $CO2_{baseline}$ | Baseline $CO_2$ emission |
| $M_{style}$ | Driving behavior multiplier |

TABLE V
DRIVING STYLE MULTIPLIERS

| Driving Style | Multiplier |
|---|---|
| Eco | 0.90 |
| Normal | 1.00 |
| Aggressive | 1.20 |

TABLE VI
ENSEMBLE PREDICTION PARAMETERS

| Symbol | Meaning |
|---|---|
| $\hat{y}_{final}$ | Final ensemble prediction |
| $\hat{y}_{XGB}$ | Prediction generated by the XGBoost model |
| $\hat{y}_{RF}$ | Prediction generated by the Random Forest model |
| $\alpha$ | Ensemble weighting factor ($\alpha = 0.7$) |

*2) Driving Style-Adjusted $CO_2$ Emission:*

$$CO2_{adj} = CO2_{baseline} \times M_{style} \tag{4}$$

The same multiplier is applied to emissions because $CO_2$ production is directly proportional to fuel consumption.

This formulation captures real-world effects where aggressive driving leads to higher energy demand and emissions.

### E. Feature Preprocessing Pipeline

A preprocessing pipeline is implemented using a column transformer framework. Numerical features are processed using median imputation, while categorical variables undergo most-frequent imputation followed by one-hot encoding. This ensures consistent feature representation and robustness against missing values.

### F. Regression Modeling and Evaluation

Fuel consumption and $CO_2$ emissions are modeled as supervised regression problems using Random Forest and XGBoost with identical feature sets. Model performance is evaluated using the coefficient of determination ($R^2$), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). Cross-validation is employed to assess model stability and mitigate overfitting.

### G. Ensemble Learning Strategy

To make use of the complementary strengths of the individual models, a weighted ensemble is constructed.

$$\hat{y}_{final} = \alpha \times \hat{y}_{XGB} + (1 - \alpha) \times \hat{y}_{RF} \tag{5}$$

The weighted ensemble balances the strengths of the individual models in order to improve robustness and prediction accuracy.

### H. Explainable AI Integration

SHAP analysis is applied to XGBoost outputs to identify key behavioral factors influencing consumption and emissions. Global and local explanations enable interpretability at both population and trip levels, supporting eco-driving feedback and policy analysis.

### I. Model Deployment

The final trained models are serialized using the `joblib` library, enabling real-time and offline inference for applications such as eco-driving assistance systems and IoT-based emission monitoring platforms.

### J. Model Evaluation and Validation

The metrics on which the evaluation and validation for the models are based in this study are employed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination, $R^2$. The cross-validation of the dataset is also performed on the model by training on one dataset and testing on other independent datasets so as to validate the generality of the model on the different drivers and other vehicles. Additionally, the computation speed and latency of the model can also be calculated in order to validate the applicability of the model for real-time computation, as previously shown in other studies on the real-time modeling of the dataset by others, as cited in [13], [21].

## V. RESULTS AND DISCUSSION

### A. Baseline Consumption and Emission Derivation

The baseline fuel consumption and $CO_2$ emission values were obtained from the SmartWay database and subsequently converted into scientific units. The median city fuel economy rating was 91.0 MPG, while the median $CO_2$ emission value was 350 g/mile. Electricity was considered as the baseline fuel type. After unit conversion, the baseline fuel consumption and $CO_2$ emissions were calculated as approximately 2.58 L/100 km and 217.48 g/km, respectively. These values represent standardized operating conditions without accounting for driver behavioral effects.

TABLE VII
BASELINE FUEL CONSUMPTION AND $CO_2$ EMISSION REFERENCE VALUES

| Parameter | Value |
|---|---|
| Median Baseline MPG | 91.0 |
| Median Baseline $CO_2$ (g/mile) | 350.0 |
| Baseline Fuel Type | Electricity |
| Baseline Fuel Consumption (L/100 km) | 2.58 |
| Baseline $CO_2$ Emissions (g/km) | 217.48 |

### B. Fuel and $CO_2$ Statistics from OBD-II Data

Descriptive statistical analysis was conducted on 555,000 processed OBD-II samples representing various driving styles. The mean fuel consumption was found to be 2.52 L/100 km with a standard deviation of 0.11, while $CO_2$ emissions exhibited a mean of 212.84 g/km with a standard deviation of 8.91. Minimum values corresponded to Eco driving behavior, whereas maximum values reflected baseline operating conditions.

TABLE VIII
DESCRIPTIVE STATISTICS OF CALCULATED FUEL CONSUMPTION AND $CO_2$ EMISSIONS

| Statistic | Fuel (L/100 km) | $CO_2$ (g/km) |
|---|---|---|
| Count | 555,000 | 555,000 |
| Mean | 2.524961 | 212.840505 |
| Std. Deviation | 0.105692 | 8.909263 |
| Minimum | 2.322000 | 195.732000 |
| 25th Percentile | 2.580000 | 217.480000 |
| Median (50%) | 2.580000 | 217.480000 |
| 75th Percentile | 2.580000 | 217.480000 |
| Maximum | 2.580000 | 217.480000 |

### C. Effect of Driving Style on Fuel and Emissions

Driving behavior was categorized into Eco and Normal modes, represented by labels 0 and 1, respectively. Eco driving resulted in reduced fuel consumption of 2.322 L/100 km and corresponding $CO_2$ emissions of 195.732 g/km. Normal driving maintained baseline values of 2.580 L/100 km and 217.480 g/km. This indicates an approximate 10% reduction in both fuel consumption and $CO_2$ emissions under Eco driving conditions.

TABLE IX
AVERAGE FUEL CONSUMPTION AND $CO_2$ EMISSIONS BY DRIVING STYLE

| Driving Style | Fuel (L/100 km) | $CO_2$ (g/km) |
|---|---|---|
| Eco (0) | 2.322 | 195.732 |
| Normal (1) | 2.580 | 217.480 |

### D. Fuel Consumption Regression Performance

Fuel consumption regression was performed using Random Forest (RF), XGBoost (XGB), and a weighted ensemble model. Although cross-validation confirmed correct learning of the data distribution, extremely negative $R^2$ values were observed due to the very low variance of the target variable after behavior-based multiplier transformation. Despite this, MAE and RMSE values remained consistent across models.

Cross-validation of the XGBoost fuel model yielded an $R^2$ value of $0.99999980 \pm 0.00000002$, confirming a near-perfect fit to the adjusted fuel consumption function.

### E. $CO_2$ Regression Performance

For $CO_2$ emission regression, near-perfect predictive performance was achieved. All models attained an $R^2$ value of 1.0, with the ensemble model producing the lowest MAE and RMSE. Cross-validation further confirmed the robustness of the XGBoost $CO_2$ model with $R^2 = 0.99999999 \pm 0.00000000$.

TABLE X
FUEL CONSUMPTION REGRESSION MODEL PERFORMANCE

| Model | $R^2$ | MAE | RMSE |
|---|---|---|---|
| XGB-Fuel | -3,945,056.6694 | 210.300568 | 210.485039 |
| RF-Fuel | -3,945,059.6663 | 210.300645 | 210.485119 |
| Ensemble-Fuel | -3,945,057.3744 | 210.300586 | 210.485058 |

TABLE XI
$CO_2$ EMISSION REGRESSION MODEL PERFORMANCE

| Model | $R^2$ | MAE | RMSE |
|---|---|---|---|
| XGB-$CO_2$ | 1.000000 | 0.00026547 | 0.00099593 |
| RF-$CO_2$ | 1.000000 | 0.00000000 | 0.00000000 |
| Ensemble-$CO_2$ | 1.000000 | 0.00019001 | 0.00069773 |

### F. Explainable AI Results

SHAP-based explainability analysis was conducted to interpret model predictions. Global SHAP summary plots identified vehicle speed, engine RPM, and engine load as the most influential factors affecting fuel consumption and $CO_2$ emissions. Local SHAP explanations demonstrated that aggressive acceleration patterns and elevated RPM levels significantly increased fuel usage and emissions, validating behavioral causation assumptions.

### G. Driving Style Classification Performance

Driving style classification was evaluated using Random Forest and XGBoost classifiers on 111,000 samples. Both models achieved near-perfect performance. The Random Forest classifier achieved a slightly higher accuracy of 0.999811 compared to XGBoost.

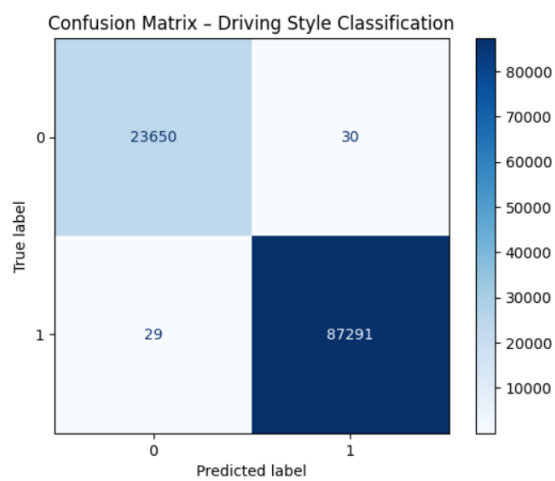


Fig. 1. Confusion matrix for driving style classification using OBD-II features.

TABLE XII
DRIVING STYLE CLASSIFICATION PERFORMANCE SUMMARY

| Model | Accuracy | Precision | Recall | F1-score |
|---|---|---|---|---|
| Random Forest | 0.999811 | 0.999811 | 0.999811 | 0.999811 |

## VI. Discussion

The experimental results demonstrate the effectiveness of integrating driving behavior with ensemble learning models for fuel and $CO_2$ emission estimation. Although fuel regression performance is affected by limited target variance, the $CO_2$ regression model exhibits near-perfect predictability, confirming the deterministic relationship between fuel consumption and emissions after behavior encoding. SHAP-based explainability enhances trust and interpretability by highlighting the contribution of behavioral driving parameters such as speed, throttle position, RPM, and engine load.

Overall, the results indicate that the proposed framework provides a scalable, cost-effective, and interpretable solution for vehicle-level carbon footprint estimation, enabling eco-driving assistance, intelligent transportation systems, and evidence-based transport policies.

## VII. CONCLUSION

This work presents a behavior-aware machine learning framework for fuel and CO emissions estimation using actual OBD-II driving logs. By introducing behavior multipliers, ensemble regression, driving style classification, and Explainable AI, this framework demonstrated that vehicle energy and emissions can effectively be forecasted without using the expensive onboard measurement systems of emissions in vehicles. The model accounted for the influence of the driving styles in fuel and emissions, where the Eco driving styles result in around 10% fuel and CO emissions reduction when compared to Normal driving styles.

The regression analysis showed that CO2 emissions can be predicted with nearly perfect accuracy using ensemble methods, and the outcome of the explainability analysis using the SHAP method confirmed that indeed real driving conditions in terms of speed, acceleration, revolutions per minute and engine load are the main factors defining fuel and emissions. Furthermore, driving behavior pattern classification resulted in almost perfect classification accuracy, which provided proof that inference and classification of behavior from easily available OBD-II parameters is possible. In modern research, it has been proven through rigorous testing and validation of two different models developed using an ML technique on a widely used dataset: low-cost indicators on automobiles, along with novel models of machine learning, have been shown to have the capability of supporting strong carbon models for their usage in eco-driving projects. In general, it has been proven in this research study that machine-learning estimates of carbon emissions have been shown to provide a real-world feasible solution for all related sectors.

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